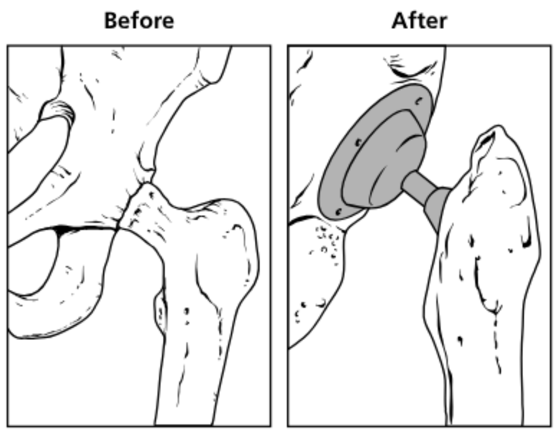
Tennessee Comprehensive
Assessment Program

TCAP

Science
Grade 7
Practice Items

TN Department of Education

A damaged hip joint can cause pain and decreased mobility. A hip replacement can restore mobility and reduce pain. During hip replacement surgery a portion of the damaged bone is removed and replaced with a man-made material.



Which of these describes a material that is **best** to use for a hip replacement?

A. a soft material that bends with the movement of the body
B. a mesh material that allows fluid in the body to pass through it
C. a hard metal material that supports the weight of the body
D. a solid dissolvable material that disappears after the body heals

A group of students investigated how cycling of matter occurs in plants and animals. They set up an investigation using both an aquatic plant and an aquatic animal. They used an indicator solution to color the water. An indicator solution will change colors depending upon how much Carbon Dioxide (CO_2) there is in the water.

Their setup and data are shown below:

Aquatic Plant and Animal Investigation

Investigation Setup	
Beaker Contents	Aquatic Plant, Snail, and Water
Exposed to Light	Yes
Initial Water Color	Green
Water Color after 1 Week	Yellow

If Carbon Dioxide turns the indicator solution yellow, which statement best explains the processes that occurred in the beaker?

A. The aquatic plant released CO_2 that the snail used to make sugars.

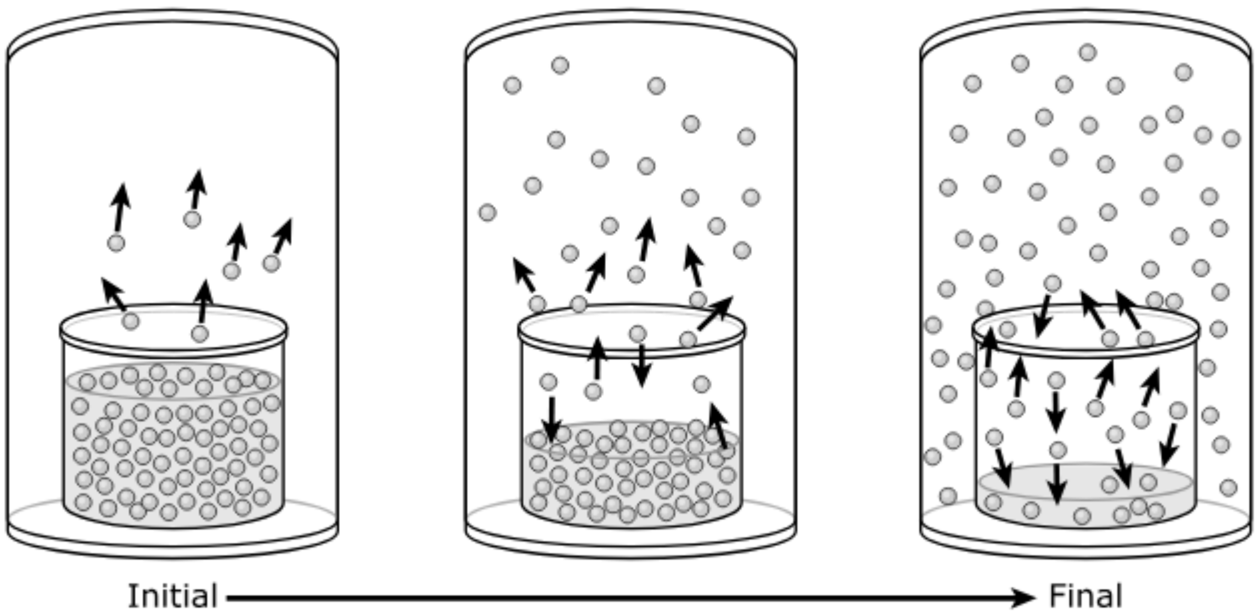
B. The water changed to green as CO_2 was consumed by the aquatic plant.

C. The snail breathed in CO_2 from the water.

D. The snail released CO_2 into the water.

The model shows a pure substance in a glass jar undergoing a change of state.

Substance Changing Its Physical State



Based on the model, the changes observed are **most likely** caused by

M. a decrease in the total mass of the substance under the glass jar without a change in temperature or pressure.

P. an increase in the temperature of the substance without a change in pressure.

R. a decrease in the temperature of the substance without a change in pressure.

S. an increase in the atmospheric pressure outside the glass jar without a change in temperature.

The questions on pages 6 - 7 refer to the passage(s) and image(s) shown.

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Frozen Balloons - Part 1

Students designed an investigation to study the effects of temperature on water and air. This was their procedure:

1. Fill two balloons with water and two balloons with air to a volume of 1.0 L.
2. Measure the circumference and the mass of each balloon.
3. Place one water-filled balloon and one air-filled balloon in a freezer set at –20°C.
4. The next day, remove the balloons from the freezer, observe any changes, and repeat the measurements in step 2.

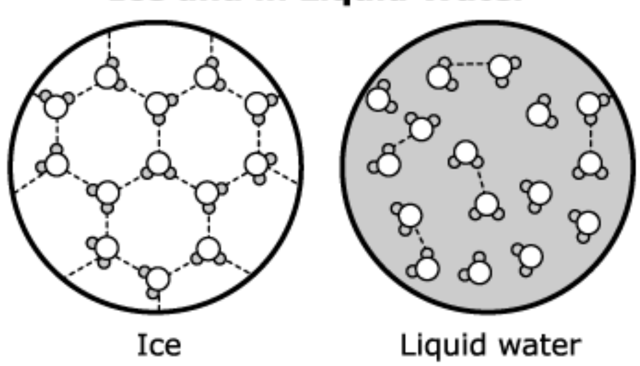
Table 1 shows the results of the water-filled balloon investigation.

Table 1. Water-Filled Balloon Investigation Results

Water-Filled Balloon	Circumference (cm)	Mass (g)
Starting room temperature	40	1000
Ending room temperature	40	1000
Before freezer	40	1000
After freezer	43	1000

The students made models of water particles in ice and liquid water. The students used the models to help them understand the results of the water-filled balloon investigation. Figure 1 shows the students' models. The dotted lines show attractions between water particles.

Figure 1. Water Particles in Ice and in Liquid Water



Frozen Balloons - Part 2

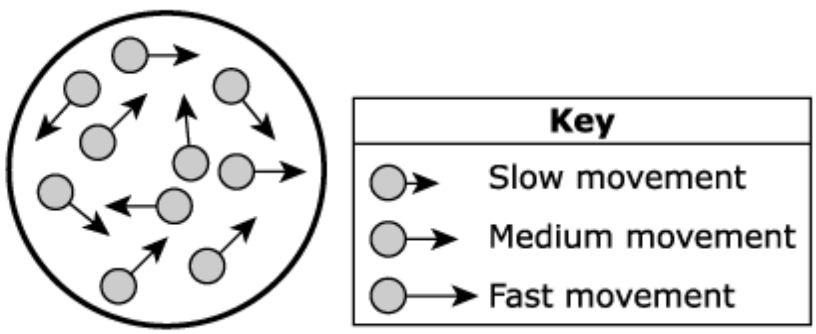
Table 2 shows the results of the air-filled balloon investigation.

Table 2. Air-Filled Balloon Investigation Results

Air-Filled Balloon	Circumference (cm)	Mass (g)
Starting room temperature	40	1.2
Ending room temperature	40	1.2
Before freezer	40	1.2
After freezer	37	1.2

The students made models of particle motion in the air-filled balloon at room temperature. The students used the models to help them understand the results of the air-filled balloon investigation. Figure 2 shows the students' models. The lengths of the arrows in the key show the speed of different particles.

Figure 2. Motion of Air Particles



1. Which statement **best** describes a change that could be made to one of the models in Figure 1 (Part 1) to show a difference between water particles in ice and water particles in liquid water?

   M. Arrows showing the movement of the water particles should be added to the liquid water model.

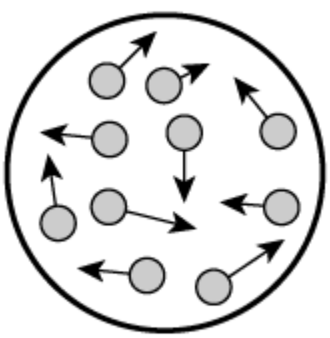
   P. Lines should be added to the ice model so that all of the water particles are connected to one another.

   R. The atoms making up the water particles in the ice model could be made larger than the atoms making up the liquid water model.

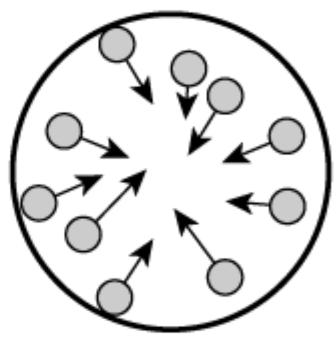
   S. The shape of the water particles making up the liquid model could be different than the shape of the water particles making up the ice model.

2. Based on Figure 2 (Part 2), which model **best** represents a change to the air particles inside the air-filled balloon after the balloon is in the freezer overnight?

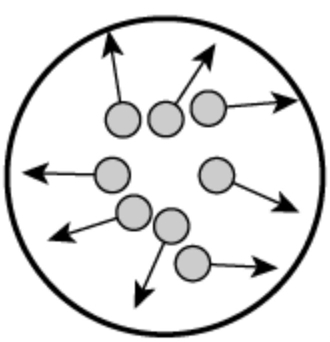
A.



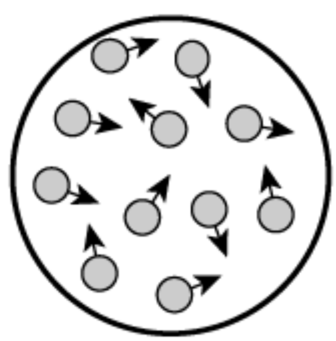
C.



B.



D.



3. Which statement correctly describes changes to the water particles and the air particles when the balloons were put in the freezer?

   M. The speeds of both the water particles and the air particles increased.

   P. The amounts of energy in both the water particles and the air particles decreased.

   R. The water particles moved closer together and the air particles moved farther apart.

   S. The pressure acting on the water particles increased and the pressure acting on the air particles decreased.

4. The students seal air inside a glass bottle and place it in the freezer. A student claims that the air pressure inside the glass bottle will decrease. Using the information in Part 1 and Part 2, which statement best supports the student's claim?

   A. Air particles will become smaller in colder temperatures.

   B. Air particles will have more room to move in colder temperatures.

   C. The air particles hit the walls of the glass bottle with less force in colder temperatures.

   D. The air particles are able to move through the sides of the glass bottle in colder temperatures.

5. The students continued to observe the balloons after removing them from the freezer. The students saw that liquid water began to appear in the frozen water balloon. The students drew a model based on the ice model in Figure 1 (Part 1) to show the melting ice. Which statement best explains how the newly created melting ice model should be different than the frozen ice model?

   M. The melting ice model should have water particles that are shaped differently.

   P. The melting ice model should have a different number of water particles.

   R. The melting ice model should show more water particles connected together.

   S. The melting ice model should show more water particles moving around.

The questions on pages 10 - 11 refer to the passage(s) and image(s) shown.

Cycles in the Tundra - Part 1

Scientists are studying the relationships between organisms in a tundra ecosystem. Table 1 describes the roles of some of the organisms that live in the tundra ecosystem.

Table 1. Roles of Organisms in the Tundra Ecosystem

Organism	Role in Ecosystem
Mosses	Producer
Shrubs	Producer
Grasses	Producer
Snow Geese	Herbivore
Lemmings	Herbivore
Weasels	Eats herbivores
Snowy Owls	Eats herbivores
Falcons	Eats all animals
Hawks	Eats all animals

Cycles in the Tundra - Part 2

The scientists also studied the relationship between the producers in Table 1 and the lemming population over a few years. They looked at how the amount of plant material consumed by herbivores affected the lemming population. Table 2 describes the data from the study. A hectare is 10,000 square meters.

Table 2. Plants Consumed and Lemming Population Sizes

Year of the Study	Percentage of Plants Consumed by Herbivores (%)	Lemming Population (number/hectare)
1	46	9
2	23	<1
3	18	<1
4	33	3
5	17	<1

1. Which model of organism relationships describes a way that energy flows through the tundra ecosystem?

   A. Shrubs → Lemmings → Weasels → Falcons

   B. Mosses → Weasels → Hawks → Snowy owls

   C. Grasses → Snow geese → Lemmings → Hawks

   D. Mosses → Shrubs → Snow geese → Snowy owls

2. Which two statements best explain how carbon moves between organisms in the tundra ecosystem?

   M. Mosses use carbon from the atmosphere.

   P. Shrubs use carbon from the biosphere.

   R. Snow Geese use carbon from the biosphere.

   S. Falcons use carbon from the atmosphere.

   T. Weasels use carbon from the atmosphere.

3. Which statement best describes the relationship between photosynthesis and cellular respiration in the tundra ecosystem?

   A. Hawks produce carbon dioxide and water, which weasels convert into sugars and oxygen.

   B. Lemmings produce sugar and water, which hawks convert into carbon dioxide and oxygen.

   C. Grasses produce sugars and oxygen, which lemmings convert into carbon dioxide and water.

   D. Mosses produce carbon dioxide and water, which weasels convert into sugars and oxygen.

4. Which two statements best describe the movement of carbon and oxygen in the tundra ecosystem?

   M. Producers take in oxygen and release carbon dioxide during photosynthesis.

   P. Producers take in carbon dioxide and release oxygen during photosynthesis.

   R. Consumers take in oxygen and release carbon dioxide during cellular respiration.

   S. Consumers take in carbon dioxide and release oxygen during cellular respiration.

   T. Producers and consumers take in carbon dioxide and release oxygen during photosynthesis.

5. Based on the information in Table 2, which statement best describes how the energy in the tundra ecosystem changes when herbivores consume different amounts of plants?

   A. Herbivores get less energy when they consume more plants.

   B. Lemmings use more energy when they consume fewer plants.

   C. The amount of energy available to other consumers changes as the number of lemmings changes.

   D. Plants produce more energy in the tundra ecosystem when fewer lemmings are in the tundra ecosystem.

Metadata – Science Items

Page Number	Item Type	Key	TN Standards
1	MC	C	7.ETS1.1
2	MC	D	7.LS1.7
3	MC	P	7.PS1.3

Frozen Balloons

Page Number	Item Type	Item Number	Key	TN Standards
4	Stimulus			
5	Stimulus			
6	MC	1	M	7.PS1.3
6	MC	2	D	7.PS1.3
7	MC	3	P	7.PS1.3
7	MC	4	C	7.PS1.3
7	MC	5	S	7.PS1.3

Cycles in the Tundra

Page Number	Item Type	Item Number	Key	TN Standards
8	Stimulus			
9	Stimulus			
10	MC	1	A	7.LS2.1
10	MS	2	M, R	7.LS2.1
10	MC	3	C	7.LS1.7
11	MS	4	P, R	7.LS1.7
11	MC	5	C	7.LS2.1

Metadata Definitions	
Item Type	MC = Multiple Choice MS = Multiple Select Stimulus = Cluster support information
Key	Correct Answer
TN Standards	Primary standard assessed in item.